Revision quiz: Chemistry

Grade 7 · IB MYP 2 · Unit 1 Chemistry's Secrets Unlocked · 20 questions · 20 marks

Name: _____ Class: _____ Date: _____

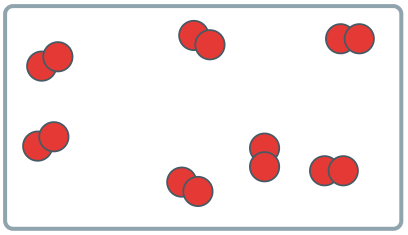
Answer every question. Each question is worth 1 mark. In questions with several parts (fill the gap, match, put in order), all parts must be right for the mark.

Total: ____ / 20

Do this quiz online (answers and explanations at the end):
https://school.taneski.com/science/g07/05-revision-quiz-chemistry/

A · PARTICLES, CHANGES AND SEPARATING

1 **The box shows the particles of a pure substance. What is it?** [1]

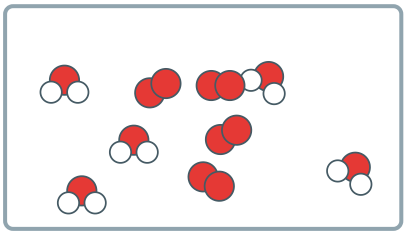


● oxygen atom

- ☐ **A** An element
- ☐ **B** A compound
- ☐ **C** A mixture
- ☐ **D** A salt

Tick one box.

2 **What does this box show?** [1]



○ hydrogen atom ● oxygen atom

- ☐ **A** An element
- ☐ **B** A compound
- ☐ **C** A mixture
- ☐ **D** Only water molecules

Tick one box.

3 **Fill in the missing words.** [1]

A compound contains two or more different (1) _____ that are (2) _____ together.

(1) choose from: types of atom / mixtures / states of matter · (2) choose from: chemically joined / only mixed / melted

4 **Is each substance an element, a compound or a mixture?** [1]

Write a letter in each box. A letter can be used more than once.

☐ 1. Oxygen gas, O_2

☐ 2. Carbon dioxide, CO_2

☐ 3. Air

☐ 4. Iron, Fe

A element

B compound

C mixture

5 **True or false? Melting ice is a chemical change.** [1]

☐ True ☐ False

6 **A student did the iron and sulfur investigation. Which result shows that a new substance was made?** [1]

Sample	Colour	Does a magnet pull out iron?
Iron	grey	yes
Sulfur	yellow	no
Iron + sulfur, mixed	grey and yellow bits	yes
Iron + sulfur, after heating	black solid	no

☐ **A** The mixture has grey and yellow bits

☐ **B** The magnet pulls the iron out of the mixture

☐ **C** After heating, the black solid is not pulled by the magnet

☐ **D** Sulfur is yellow

Tick one box.

7 **You have a mixture of sand and salt. Put the steps in order to get dry salt.** [1]

Number the steps 1 to 4 in the right order.

☐ **A** Heat the salty water in an evaporating dish

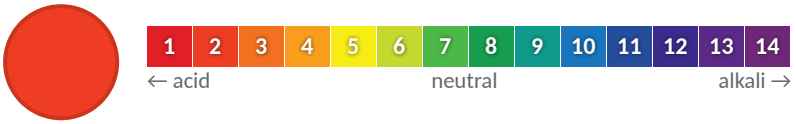
☐ **B** Add water and stir, so the salt dissolves

☐ **C** Salt crystals are left when the water has evaporated

☐ **D** Filter: the sand stays in the filter paper

B · ACIDS, ALKALIS AND THE PH SCALE

8 **Universal indicator turns a liquid this colour. Use the chart: what is the liquid?** [1]



☐ **A** pH 2 – very acidic

☐ **B** pH 7 – neutral

☐ **C** pH 9 – slightly alkaline

☐ **D** pH 13 – very alkaline

Tick one box.

9 **Look at the results. Which liquid is slightly alkaline?** [1]

Liquid	Universal indicator colour	pH
Lemon juice	red	2
Distilled water	green	7
Hand soap	blue	10
Oven cleaner	purple	13

- ☐ **A** Lemon juice
- ☐ **B** Distilled water
- ☐ **C** Hand soap
- ☐ **D** Oven cleaner

Tick one box.

10 **Which of these is NOT a property of acids?** [1]

- ☐ **A** They have a pH below 7
- ☐ **B** They turn blue litmus red
- ☐ **C** They feel soapy
- ☐ **D** Food acids taste sour

Tick one box.

11 **Match each test to the colour you would see.** [1]

Write a letter in each box. A letter can be used more than once.

- ☐ 1. Litmus in an acid
- ☐ 2. Litmus in an alkali
- ☐ 3. Universal indicator in pure water
- ☐ 4. Universal indicator in oven cleaner (pH 13)

A red
B blue
C green
D purple
E orange

12 **Lots of water is added to vinegar (pH 3). What could the pH be now?** [1]

- ☐ **A** pH 1
- ☐ **B** pH 3
- ☐ **C** pH 5
- ☐ **D** pH 9

Tick one box.

13 **True or false? A dilute acid has fewer acid particles in each drop than a concentrated acid.** [1]

☐ True ☐ False

14 **Fill in the missing words.** [1]

Burning coal in power stations releases (1) _____. It dissolves in the water in clouds and makes (2) _____, which harms lakes, trees and buildings.

(1) choose from: sulfur dioxide / oxygen / water vapour · (2) choose from: acid rain / alkaline rain / pure rain

C · NEUTRALISATION AND SALTS

15 **Fill in the missing words.** [1]

Complete the word equation: acid + alkali → (1) _____ + (2) _____

(1) choose from: sugar / a salt / hydrogen · (2) choose from: water / oxygen / carbon dioxide

16 **Hydrochloric acid is added drop by drop to sodium hydroxide with universal indicator, until it is neutral. Which colours do you see, in order?** [1]

- ☐ **A** purple → blue → green
- ☐ **B** green → blue → purple
- ☐ **C** red → orange → green
- ☐ **D** green → red → purple

Tick one box.

17 **Put the steps for making table salt (sodium chloride) in order.** [1]

Number the steps 1 to 4 in the right order.

- ☐ **A** Heat gently until most of the water has evaporated: salt crystals form
- ☐ **B** Add hydrochloric acid drop by drop until the indicator is green
- ☐ **C** Measure sodium hydroxide into a beaker and add universal indicator
- ☐ **D** Add charcoal, then filter the solution into an evaporating dish

18 **Which salt ending comes from each acid?** [1]

Write a letter in each box. A letter can be used more than once.

☐ 1. Hydrochloric acid	**A** …chloride
☐ 2. Sulfuric acid	**B** …sulfate
☐ 3. Nitric acid	**C** …nitrate
	D …sulfide

19 **Complete: nitric acid + potassium hydroxide → ..... + water** [1]

- ☐ **A** potassium nitrate
- ☐ **B** potassium chloride
- ☐ **C** nitric potassium
- ☐ **D** potassium sulfate

Tick one box.

20 **Match each problem to what neutralises it.** [1]

Write a letter in each box. A letter can be used more than once.

☐ 1. Bee sting (acidic)	**A** baking soda
☐ 2. Wasp sting (alkaline)	**B** vinegar
☐ 3. Too much stomach acid	**C** indigestion tablet
☐ 4. Acidic soil on a farm	**D** lime
	E lemon juice